

DUNGEON QUEST

WEEK 7 REPORT

Advanced Game Version Development

1. Name of the Experiment

Development of Boss Battle, Quest System and Audio-Visual Improvements in Dungeon Quest RPG

2. Objective

The main objective of Week 7 was to improve the existing Dungeon Quest RPG by adding advanced gameplay features. A boss battle system and quest system were introduced to make the game more challenging and meaningful. The visual environment was also improved by adding a grass field, rocks, and character-like designs. Short sound effects were added for fighting and other important actions.

3. Introduction

Dungeon Quest RPG is a simple 2D role-playing game developed using Java Swing and AWT. In the previous weeks, the basic player movement, enemy AI, combat, weapon, XP, and level systems were developed.

In Week 7, these existing features were improved and new elements were added. The main focus was to make the game feel more like a complete RPG rather than a basic prototype. For this purpose, a powerful boss, a simple quest system, improved graphics, obstacles, and sound effects were added.

4. Development Environment

The game was developed using the following tools and technologies:

- Programming Language: Java
- GUI: Java Swing and AWT
- IDE: Visual Studio Code
- Platform: Windows
- Game Type: 2D RPG
- Graphics: Java Graphics2D
- Audio: Java Sound API

The main files used in this version are Main.java, GameFrame.java, GamePanel.java, Player.java, Enemy.java, Dungeon.java, Boss.java, Quest.java, and AudioManager.java.

5. Boss Battle System

Main boss features include:

- High health
- Faster movement
- Stronger attack
- Rage effect
- Boss health bar
- Extra XP reward
- Special skill cooldown system

6. Quest System

A simple quest system was introduced through the Quest.java class.

The quest gives the player a clear objective during the game. The objectives change according to the player's progress.

The quest follows this sequence:

Find the Sword → Defeat 3 Enemies → Defeat the Boss → Quest Complete

At the beginning, the player needs to find the sword. After collecting it, the objective changes to defeating three normal enemies. Once the three enemies are defeated, the player is instructed to fight the boss.

The current objective is displayed on the game screen. This makes it easier for the player to understand what to do next.

7. Audio Improvement

The previous continuous background sound was removed because it created unnecessary noise during gameplay.

Instead, short sound effects were added for important actions. A small sound is played when the player attacks an enemy. Different sounds are also used when the player collects the weapon, defeats a normal enemy, or defeats the boss.

This makes the game quieter while still providing audio feedback.

The main sound effects are:

- Weapon collection sound
- Fighting or attack sound
- Enemy defeat sound
- Boss defeat sound

The sounds were generated using Java's Sound API, so external sound files were not required.

8. Visual Improvement

The visual design of the game was improved considerably.

Previously, the game had a simple black background and square-shaped player and enemies. In this version, the black background was replaced with a green grass field.

Small grass details were added to make the environment look less empty. Rock objects were also placed around the map. These rocks work as obstacles and prevent the player from simply moving through them.

The player and normal enemies were redesigned using circles and rounded shapes to give them a simple character or doll-like appearance. The boss was given a larger and different design so that it can be easily identified.

9. Combat and Game Progression

The existing combat and progression systems from the previous weeks were kept functional.

The player can collect the sword and increase attack power. By pressing SPACE, the player can attack nearby enemies.

When an enemy is defeated, the player receives XP. Normal enemies provide 50 XP, while the boss provides 100 XP.

When the required XP is reached, the player's level increases and attack power is improved.

The basic controls are:

W = Move Up

S = Move Down

A = Move Left

D = Move Right

SPACE = Attack

U = Upgrade Weapon

10. Implementation Process

The Week 7 development was completed in several steps.

First, the Boss.java class was created and the boss was added to the dungeon. Then the Quest.java class was added to control the objectives and quest progress.

After that, GamePanel.java was updated to handle the boss, quest, improved graphics, rocks, and sound effects.

The game background was changed to a grass field and several rock objects were added. The player, enemies, and boss were also given improved character-like designs.

Finally, the sound system was modified to remove the continuous background noise and use short sound effects during important game events.

11. Testing and Result

After completing the implementation, the major features were tested.

The player was successfully moved using the WASD keys. The sword could be collected and the attack power increased. Normal enemies could be attacked and defeated, and XP was added correctly.

The quest objective changed as the player completed different tasks. After defeating the normal enemies, the boss became the final challenge.

The boss had higher health and stronger attacks than normal enemies. The boss health bar also displayed its remaining health.

The grass background, rocks, character designs, and sound effects worked together to improve the overall game experience.

12. Discussion

The main purpose of Week 7 was to improve the gameplay and appearance of the existing RPG. The boss system added a final challenge, while the quest system gave the player a clear goal.

The visual changes also made the game more attractive compared to the previous simple black background and square objects. The addition of rocks made the environment more interactive.

Removing the continuous background sound was another useful improvement. Short fighting and event sounds provide feedback without making the game noisy.

Although the current version is still a simple 2D RPG, it now has the basic structure of a more complete game. In the future, more detailed character graphics, real background music, different weapons, multiple bosses, and more advanced quests can be added.

13. Conclusion

In Week 7, the Dungeon Quest RPG was successfully upgraded to an advanced version. A boss battle system, quest system, improved game environment, character-like graphics, obstacles, and short sound effects were added.

The new features work together with the previously developed combat, weapon, XP, and level systems. As a result, the game became more challenging, interactive, and visually appealing.

Overall, the Week 7 objective of developing an Advanced Game Version was successfully achieved.